

Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-18.jpg, we are tasked with evaluating the following limit:

$$L = \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} - \sqrt{1 + \sqrt{2}}}{x^2}$$

1. Successive Rationalization

Because direct substitution of $x = 0$ yields the indeterminate form $\frac{0}{0}$, we can evaluate this limit by repeatedly multiplying the numerator and the denominator by the conjugate of the radical expressions to eliminate the nested square roots. Let the expression inside the limit be $f(x)$.

First, multiply by the conjugate of the outermost difference, $\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} + \sqrt{1 + \sqrt{2}}$:

$$f(x) = \frac{\left(1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}\right) - (1 + \sqrt{2})}{x^2 \left(\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} + \sqrt{1 + \sqrt{2}}\right)}$$

Simplifying the numerator:

$$f(x) = \frac{\sqrt{1 + \sqrt{1 + x \sin(x)}} - \sqrt{2}}{x^2 \left(\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} + \sqrt{1 + \sqrt{2}}\right)}$$

Next, multiply by the conjugate of the new numerator, $\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2}$:

$$\begin{aligned} f(x) &= \frac{\left(1 + \sqrt{1 + x \sin(x)}\right) - 2}{x^2 \left(\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2}\right)} \\ &\quad \times \frac{1}{\left(\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2}\right)} \end{aligned}$$

Which simplifies to:

$$\begin{aligned} f(x) &= \frac{\sqrt{1 + x \sin(x)} - 1}{x^2 \left(\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2}\right)} \\ &\quad \times \frac{1}{\left(\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2}\right)} \end{aligned}$$

Finally, multiply by the conjugate of the remaining numerator, $\sqrt{1 + x \sin(x)} + 1$:

$$f(x) = \frac{(1 + x \sin(x)) - 1}{x^2 \left(\sqrt{1 + \sqrt{\dots}} + \sqrt{1 + \sqrt{2}} \right) \left(\sqrt{1 + \sqrt{\dots}} + \sqrt{2} \right)} \times \frac{1}{\left(\sqrt{1 + x \sin(x)} + 1 \right)}$$

Simplifying the numerator gives our fully rationalized expression:

$$f(x) = \frac{x \sin(x)}{x^2 \left(\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} + \sqrt{1 + \sqrt{2}} \right)} \times \frac{1}{\left(\sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2} \right) \left(\sqrt{1 + x \sin(x)} + 1 \right)}$$

2. Applying the Limit

We can now factor the expression to isolate the known limit $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$:

$$L = \lim_{x \rightarrow 0} \left[\frac{\sin(x)}{x} \cdot \frac{1}{D_1(x)D_2(x)D_3(x)} \right]$$

where the denominator terms $D_i(x)$ are the conjugates we multiplied by:

$$\begin{aligned} D_1(x) &= \sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} + \sqrt{1 + \sqrt{2}} \\ D_2(x) &= \sqrt{1 + \sqrt{1 + x \sin(x)}} + \sqrt{2} \\ D_3(x) &= \sqrt{1 + x \sin(x)} + 1 \end{aligned}$$

As $x \rightarrow 0$, the term $x \sin(x) \rightarrow 0$. Evaluating the limits of the denominator factors via direct substitution:

$$\begin{aligned} \lim_{x \rightarrow 0} D_1(x) &= \sqrt{1 + \sqrt{1 + \sqrt{1 + 0}}} + \sqrt{1 + \sqrt{2}} = 2\sqrt{1 + \sqrt{2}} \\ \lim_{x \rightarrow 0} D_2(x) &= \sqrt{1 + \sqrt{1 + 0}} + \sqrt{2} = 2\sqrt{2} \\ \lim_{x \rightarrow 0} D_3(x) &= \sqrt{1 + 0} + 1 = 2 \end{aligned}$$

Conclusion

Substituting these limits back into our overarching equation yields:

$$\begin{aligned} L &= 1 \cdot \frac{1}{\left(2\sqrt{1 + \sqrt{2}} \right) \cdot (2\sqrt{2}) \cdot (2)} \\ L &= \frac{1}{8\sqrt{2}\sqrt{1 + \sqrt{2}}} \end{aligned}$$

By multiplying the numerator and denominator by $\sqrt{2}$, we can also express this in a slightly cleaner rationalized form:

$$L = \frac{\sqrt{2}}{16\sqrt{1+\sqrt{2}}}$$

Therefore, the evaluated limit is:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + \sqrt{1 + x \sin(x)}}} - \sqrt{1 + \sqrt{2}}}{x^2} = \frac{1}{8\sqrt{2 + 2\sqrt{2}}}$$